

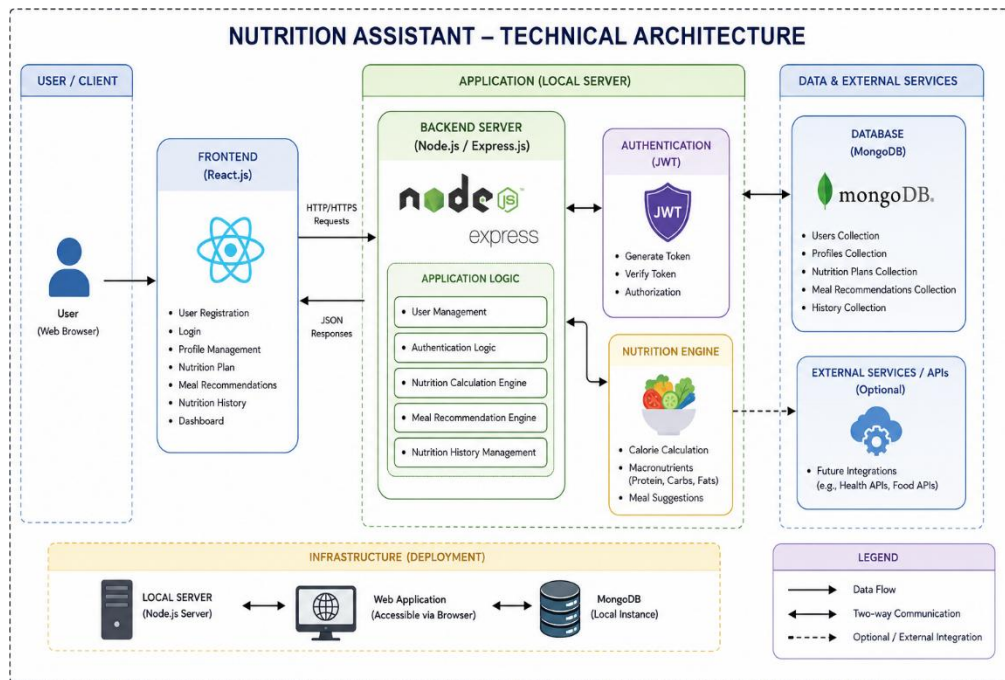
Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	13 July 2026
Team ID	SWTID-2026-8698
Project Name	Nutrition Assistant
Maximum Marks	4 Marks

Technical Architecture:

Example: Personalized Nutrition Planning System using MERN Stack



Guidelines:

- User registration, login, nutrition plan generation, meal recommendations, and nutrition history modules.
- React frontend, Express.js backend, and MongoDB database architecture.
- Local server infrastructure with backend and database communication.
- JWT authentication for secure user access and authorization.
- User profile data and nutrition plan storage in MongoDB.
- HTTP/HTTPS communication between frontend and backend.
- Axios for API requests and JSON data exchange.
- Nutrition calculation logic for calories, protein, carbohydrates, fats, and meal suggestions.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web interface for registration, login, and nutrition plan generation	React.js, HTML, CSS, Tailwind CSS
2.	Application Logic-1	User registration, login, and authentication	Node.js, Express.js
3.	Application Logic-2	Nutrition calculation and meal recommendation logic	JavaScript (Node.js)
4.	Application Logic-3	JWT authentication and authorization	JWT, bcryptjs
5.	Database	Stores user details and nutrition plans	MongoDB, Mongoose
6.	Cloud Database	Not Applicable (Local MongoDB)	–
7.	File Storage	Stores application files locally	Local File System
8.	External API-1	Not Used	–
9.	External API-2	Not Used	–
10.	Machine Learning Model	Not Used	–
11.	Infrastructure (Server / Cloud)	Local deployment using Node.js server	Local Server

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frontend and backend built using open-source technologies.	React.js, Node.js, Express.js, MongoDB
2.	Security Implementations	Secure authentication and password encryption	JWT, bcryptjs
3.	Scalable Architecture	Three-tier architecture supporting future enhancements	MERN Stack

S.No	Characteristics	Description	Technology
4.	Availability	Accessible through a local web server	Node.js Local Server
5.	Performance	Fast API responses and efficient database queries	Express.js, MongoDB